

Mengdie Huang (Maggie)

PostDoctoral Research Associate, Purdue University

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RESEARCH INTEREST

My research interests lie in security and privacy in machine learning, networking, and multimedia. I have been focusing on the robustness of deep neural networks and large (vision) language models against adversarial attacks and distribution shifts, with particular emphasis on robust training and certified defenses.

I have a strong background in deep learning, including techniques like manifold learning, randomized smoothing, transfer learning, contrastive learning, knowledge distillation, parameter-efficient fine-tuning, unlearning, and membership inference, applied to prediction and generation tasks in image, text, and network traffic domains.

EDUCATION

Purdue University , West Lafayette, IN <i>Visiting Ph.D. in Computer Science</i> Advisor: Dr. Elisa Bertino	Sep 2022 – Jun 2025
Xidian University , Xi'an, China Thesis: <i>Research on Key Techniques for Adversarial Robustness of Deep Neural Networks.</i> Advisor: Dr. Xiaofeng Chen	Sep 2019 – Jun 2025
Communication University of China , Beijing, China <i>Master in Electronics and Communication Engineering</i> <i>Bachelor in Radio and Television Editing</i>	Sep 2017 – Jun 2019
Huai'an University , Huai'an, China <i>Bachelor in Communication Engineering</i>	Sep 2013 – Jun 2017

PUBLICATIONS

Papers Published

- [1] **Mengdie Huang**, Yingjun Lin, Ninghui Li, Xiaofeng Chen, Elisa Bertino[✉]. TriXfer: Triple-Cross Transfer Attack on Vision Transformer-Adapted Downstream Classifiers[J]. *IEEE Transactions on Dependable and Secure Computing (TDSC)*, 2026. (CCF A)
- [2] **Mengdie Huang**, Yingjun Lin, Ninghui Li, Xiaofeng Chen, Elisa Bertino[✉]. CARD: Robustness-Preserving Transfer Learning for Network Intrusion Detection via Contrastive Adversarial Representation Distillation[J]. *IEEE Transactions on Dependable and Secure Computing (TDSC)*, 2025, pp. 5134-5151. (CCF A)
- [3] **Mengdie Huang**, Yingjun Lin, Xiaofeng Chen, Elisa Bertino[✉]. Dimensional Robustness Certification for Deep Neural Networks in Network Intrusion Detection Systems[J]. *ACM Transactions on Privacy and Security (TOPS)*, 2025, pp. 1-33. (CCF B)
- [4] **Mengdie Huang**, Yingjun Lin, Xiaofeng Chen[✉], Elisa Bertino. MARS: Robustness Certification for Deep Network Intrusion Detectors via Multi-Order Adaptive Randomized Smoothing[C]. *IEEE International Conference on Trust, Security, and Privacy in Computing and Communications (TrustCom)*, 2024, pp. 767-774. (**Best Paper**) (CCF C)
- [5] **Mengdie Huang**[✉], Hyunwoo Lee, Ashish Kundu, Xiaofeng Chen, Anand Mudgerikar, Ninghui Li, Elisa Bertino. ARIoTEDef: Adversarially Robust IoT Early Defense System based on Self-evolution against Multi-step Attacks[J]. *ACM Transactions on Internet of Things (TIOT)*, 2024, 5(3): 1-34.
- [6] **Mengdie Huang**, Yi Xie, Xiaofeng Chen[✉], Jin Li, Changyu Dong, Zheli Liu, Willy Susilo. Boost Off/On-Manifold Adversarial Robustness for Deep Learning with Latent Representation Mixup[C]. *ACM Asia Confer-*

ence on Computer and Communications Security (AsiaCCS), 2023, pp. 716-730. (CCF C)

- [7] **Mengdie Huang**, Cheng Yang[✉], Hao Li, Jian Shen. Sparse Selective Encryption for HEVC 4K Video Using Spatial Error Spread[J]. *Journal of Internet Technology (JIT)*, 2019, 20(5): 1589-1600.
- [8] **Mengdie Huang**, Cheng Yang[✉], Yuan Zhang. Selective Encryption of H.264/AVC based on Block Weight Model[C]. *IEEE International Conference on Communication Technology (ICCT)*, 2018, pp. 1368-1373.
- [9] Elisa Bertino[✉], Hyunwoo Lee, **Mengdie Huang**, Charalampos Katsis, Zilin Shen, Bruno Ribeiro, Daniel De Mello, Ashish Kundu. A Pro-Active Defense Framework for IoT Systems[C]. *IEEE International Conference on Collaboration and Internet Computing (CIC)*, 2023, pp. 125-132.
- [10] Yi Xie, **Mengdie Huang**, Xiaoyu Zhang, Changyu Dong, Willy Susilo, Xiaofeng Chen[✉]. GAME: Generative-based Adaptive Model Extraction Attack[C]. *European Symposium on Research in Computer Security (ESORICS)*, 2022, pp. 570-588. (CCF B)
- [11] Shanyue Bu[✉], **Mengdie Huang**, Kun Yu. An Effective Scheme for Provable Data Possession[C]. *International Conference on Intelligent Control and Computer Application (ICCA)*, 2016, pp. 344-348.

Papers Under Review

- [1] Chen Peng*, **Mengdie Huang***[✉], Ninghui Li, Elisa Bertino. Local Knowledge for Global Impact: Targeted Evasion Attack Against Vertical Federated Learning, 2026. (* joint first authors.)
- [2] MD Shamsul Kaonain*, **Mengdie Huang***[✉], Imtiaz Karim, Elisa Bertino. CARE: Robust Transformer-based Network Intrusion Detection via Consistency-aware Attention-Representation Enhancement, 2026. (* joint first authors)
- [3] Mir Imtiaz Mostafiz*, **Mengdie Huang***[✉], Imtiaz Karim, Elisa Bertino. GRAFT: Adversarially Robust Transfer Learning for Graphs Under Structural Data Scarcity, 2026. (* joint first authors)
- [4] Yani Liu, Feng Zhang[✉], Yu Zhang, **Mengdie Huang**, Elisa Bertino, Xiaoyong Du. EntroDelta: An Error-Bounded, Data-Free Compression Framework for Massive Model Storage. 2026.
- [5] Zhongkui Ma, **Mengdie Huang**, Guangdong Bai[✉], Elisa Bertino, Zihan Wang. Local Nonlinear Substitution for Bounding Activation Functions, 2026.

Books

- [1] Xiaofeng Chen[✉], **Mengdie Huang**, etc. Cloud Computing Security (2nd Edition)[B]. *Science Press. China Science Publishing & Media Co., Ltd.* 2023.
- [2] Xiaofeng Chen[✉], **Mengdie Huang**, etc. Searchable Encryption[B]. *Science Press. China Science Publishing & Media Co., Ltd.* 2024.

Employment

Purdue University , West Lafayette, IN	July 2025 – Present
<i>PostDoctoral Research Associate</i> in the Department of Computer Science	
Advisor: Dr. Elisa Bertino and Dr. Ninghui Li	
Research: Robustness of large language and vision foundation models.	
Purdue University , West Lafayette, IN	Sep 2022 - June 2025
<i>Visiting Scholar (PhD Student)</i> in the Department of Computer Science	
Advisor: Dr. Elisa Bertino	
Research: Robustness of deep learning-based network traffic analysis systems.	
Xidian University , Xi'an, China	Sep 2019 – Aug 2022
<i>Research Assistant</i> in the School of Cyber Engineering	
Advisor: Dr. Xiaofeng Chen	

Research: Robustness of deep learning-based image classification systems Xidian University , Xi'an, China <i>Teaching Assistant</i> in the School of Cyber Engineering Instructor: Dr. Mirosław Kutylowski in <i>Distributed Computing</i>	Jul 2021 - Jul 2022
Xidian University , Xi'an, China <i>Teaching Assistant</i> in the School of Cyber Engineering Instructor: Dr. Jianfeng Wang in <i>Probability Theory and Mathematical Statistics</i>	Sep 2020 - Jan 2021
National Radio and Television Administration , Beijing, China <i>Research Assistant</i> in <i>Academy of Broadcasting Science</i> Research: Secure video coding standards.	May 2019 - Jul 2019
ByteDance Ltd , Beijing, China <i>Internship</i> in <i>Commercialization Department</i>	Aug 2018 - Sep 2018

GRANT WRITING EXPERIENCE

Author of the NSF proposal (as postdoc) , Purdue – West Lafayette, IN <i>Model Robustness, Ownership, and Privacy Preserving in Transfer learning</i>	Sep 2026
Author of the proposal (as student) , Purdue – West Lafayette, IN <i>Detection of GenAI Generated Malware Variants and Sandbox Evasion using GenAI</i> , Awarded \$150,000 by Cisco Research	Mar 2023

ACADEMIC SERVICE

Journal Reviewer

- IEEE Transactions on Dependable and Secure Computing (TDSC): 2025 - Present
- IEEE Transactions on Networking (TNET): 2025 - Present
- IEEE Transactions on Parallel and Distributed Systems (TPDS): 2025-Present
- IEEE Transactions on Information Forensics & Security (TIFS): 2024 - Present
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS): 2024 - Present
- ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM): 2024 - Present
- Computers and Electrical Engineering (CEE): 2024 - Present
- Computing and Informatics (CAI): 2024 - Present
- American Journal of Artificial Intelligence (AJAI): 2024 - Present
- ACM Computing Surveys: 2023 - Present
- IEEE Transactions on Knowledge and Data Engineering (TKDE): 2022 - Present
- Telecommunication Systems (TELS): 2021 - Present
- Computer Standards & Interfaces (CSI): 2021 - Present
- Connection Science: 2020 - Present
- IEEE Transactions on Circuits and Systems for Video Technology (TCSVT): 2019 - Present

PC Member

- IEEE International Conference on Parallel and Distributed Systems (ICPADS): 2026
 - Track: Agentic Design for System and Network

Conference Reviewer

- International Conference on Data Security and Privacy Protection (DSPP): 2024, 2025
- Annual Computer Security Applications Conference (ACSAC): 2024
- European Symposium on Research in Computer Security (ESORICS): 2021, 2024, 2025
- ACM Conference on Data and Application Security and Privacy (CODASPY): 2023, 2024, 2025

- International Conference on Machine Learning for Cyber Security (ML4CS): 2022
- Information Security Conference (ISC): 2022
- International Conference on Information and Communications Security (ICICS): 2021

Conference SubReviewer

- IEEE Symposium on Security and Privacy (S&P): 2025, 2026
- ACM Conference on Computer and Communications Security (CCS): 2025
- USENIX Security (USENIX Security Symposium): 2026

Artifact Evaluation Committee Member

- ACM Conference on Computer and Communications Security (CCS): 2025, 2026
- Network and Distributed System Security Symposium (NDSS): 2026

PROJECTS

Pro-Active Attack Management for Edge Computing Security , West Lafayette, IN	Nov 2022 - Dec 2023
Developed a robust LSTM-based network intrusion detection system to defend against cyber chain attacks.	
Sponsor: Cisco Systems, Inc.	
Universal and Efficient 4K Video Protection Technology Supporting Coding Standard Extension , Beijing, China	Jun 2018 - Jun 2019
Developed A Secure and Efficient 4K HEVC Video Encoding Framework.	
Sponsor: National College Students Innovation and Entrepreneurship Training Program.	

MENTORING

Chen Peng , Ph.D. Student, Purdue University	Jun 2026 - Present
Research: Machine learning security.	
Md Shamsul Kaonain , Ph.D. Student, Purdue University	Jun 2025 - Present
Research: Network Traffic Analysis with deep learning.	
Yani Liu , Visiting Ph.D. Student, Purdue University	Mar 2025 - Present
Research: Data compression in machine learning.	
Mir Imtiaz Mostafiz , Ph.D. Student, Purdue University	Jan 2025 - Present
Research: Robustness of graph neural networks.	
Jason Hu , Undergraduate Student, Purdue University	Feb 2024 - Present
Research: Robust Transfer Learning.	
Yingjun Lin , Undergraduate Student, Purdue University	Nov 2022 - Dec 2023
Research: Adversarial attack and robustness in deep learning.	
Weihao Liu , Undergraduate student, Xi'an University of Posts and Telecommunications	Dec 2020 - Jun 2021
Research: Iterative adversarial attack with projected gradient descent.	
Mengchen Wang , Undergraduate student, Xi'an University of Posts and Telecommunications	Dec 2020 - Jun 2021
Research: Adversarial attack with Jacobian-based Saliency Map.	
Wei Li , Undergraduate student, Xidian University	Dec 2019 - Jun 2020
Research: Adversarial attack with Fast Gradient Sign Method.	
Fengdong Li , Undergraduate student, Communication University of China	Sep 2018 - Jun 2019
Research: Selective Encryption for HEVC Video.	
Yujie Wang , Undergraduate student, Communication University of China	Sep 2018 - Jun 2019
Research: Selective Encryption for HEVC Video.	

TECHNOLOGY CONTEST AWARDS

Blue Bridge Cup National Software Development Technology Competition Jiangsu Province Second Prize, China	Mar 2016
National College Student Innovation and Entrepreneurship Competition Jiangsu Province Third Prize, China	Sep 2015
Challenge Cup National College Students Extracurricular & Academic Contest Jiangsu Province First Prize, China	Jun 2015

SKILLS

Programming: Python, C + +, HTML, Java, SQL, MATLAB

Tools: Pytorch, Sklearn, Keras, HuggingFace, VScode, etc.

Language: English, Chinese

Certificate: National Computer Rank Examination Grade 4 (NCRE-4), Network Engineer

SELECTED AWARDS & HONORS

- Outstanding Doctoral Student Awarded by Xidian University	Sep 2019 - Sep 2022
- Xidian University Doctoral Academic Scholarship for Four Consecutive Years	Sep 2019 - Sep 2022
- Merit Graduate Student Awarded by Communication University of China	Jun 2019
- Communication University of China Graduate Academic Scholarship	Sep 2017 - Sep 2018
- Integrated Communication Social Scholarship Supported by Ze Media	Sep 2018

REFERENCES

Dr. Elisa Bertino, Department of Computer Science, West Lafayette, IN	Purdue University
Dr. Ninghui Li, Department of Computer Science, West Lafayette, IN	Purdue University
Dr. Xiaofeng Chen, School of Cyber Engineering, Xi'an China	Xidian University